

Laboratory Operations

Calculations, Quality Control, and Statistics

1. **How many grams of sodium hydroxide (NaOH) are required to prepare 150.0 mL of a 5.0% weight/volume (w/v) solution?**

A. 1.5 g
B. 4.0 g
C. 7.5 g
D. 15.0 g

Laboratory Operations/Calculate/Solutions/2

2. **How many milliliters of glacial acetic acid are needed to prepare 2.0 L of 10.0% v/v acetic acid?**

A. 10.0 mL
B. 20.0 mL
C. 100.0 mL
D. 200.0 mL

Laboratory Operations/Calculate/Solutions/2

3. **A biuret reagent requires preparation of a stock solution containing 9.6 g of copper II sulfate (CuSO_4) per liter. How many grams of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ are needed to prepare 1.0 L of the stock solution?**

Atomic weights: H = 1.0; Cu = 63.6; O = 16.0; S = 32.1

A. 5.4 g
B. 6.1 g
C. 15.0 g
D. 17.0 g

Laboratory Operations/Calculate/Reagent preparation/2

Answers to Questions 1–3

1. **C** A percent solution expressed in w/v refers to grams of solute per 100.0 mL of solution. To calculate, multiply the percentage (as grams) by the volume needed (mL), and then divide by 100.0 (mL).

$$(5.0 \text{ g} \times 150.0 \text{ mL}) \div 100.0 \text{ mL} = 7.5 \text{ g}$$

To prepare the solution, weigh 7.5 g of NaOH pellets, and add to a 150.0-mL volumetric flask. Add sufficient deionized H_2O to dissolve the NaOH. After the solution cools, add deionized H_2O to the 150.0 mL line on the flask, and mix again.

2. **D** The expression percent v/v refers to the volume of one liquid in milliliters present in 100.0 mL of solution. To calculate, multiply the percentage (as mL) by the volume required (mL), then divide by 100 (mL).

$$(10.0 \text{ mL} \times 2000.0 \text{ mL}) \div 100.0 \text{ mL} = 200.0 \text{ mL}$$

To prepare 2.0 L of a 10.0% v/v solution of acetic acid, add approximately 1.0 L of deionized H₂O to a 2.0 L volumetric flask. Add 200.0 mL of glacial acetic acid, and mix. Then, add sufficient deionized H₂O to bring the meniscus to the 2.0 L line, and mix again.

3. **C** Determine the mass of CuSO₄ • 5H₂O containing 9.6 g of anhydrous CuSO₄. First, calculate the percentage of CuSO₄ in the hydrate, and then divide the amount needed (9.6 g) by the percentage.

$$\% \text{ CuSO}_4 = \text{molecular weight (MW) CuSO}_4 \div \text{MW CuSO}_4 \cdot 5\text{H}_2\text{O} \times 100 = (159.7 \div 249.7) \times 100$$

$$= 63.96\%$$

$$\text{Grams CuSO}_4 \cdot 5\text{H}_2\text{O} = 9.6 \text{ g} \div 0.6396 = 15.0 \text{ g}$$

A convenient formula to use is:

$$g \text{ hydrate} = (\text{MW hydrate} \div \text{MW anhydrous salt}) \times g \text{ anhydrous salt}$$

4. **How many milliliters of nitric acid (HNO₃) (purity 68.0%, specific gravity 1.42) are needed to prepare 1.0 L of a 2.0 N solution?**

Atomic weights: H = 1.0; N = 14.0; O = 16.0

- A. 89.5 mL
- B. 126.0 mL
- C. 130.5 mL
- D. 180.0 mL

Laboratory Operations/Calculate/Reagent preparation/2

5. **Convert 10.0 milligrams per deciliter (mg/dL) calcium (atomic weight = 40.1) to the International System of Units (SI).**

- A. 0.25
- B. 0.40
- C. 2.5
- D. 0.4

Laboratory Operations/Calculate/SI unit conversion/2

6. **Convert 2.0 milliequivalents per liter (mEq/L) magnesium (atomic weight = 24.3) to mg/dL.**

- A. 0.8 mg/dL
- B. 1.2 mg/dL
- C. 2.4 mg/dL
- D. 4.9 mg/dL

Laboratory Operations/Calculate/Unit conversion/2

7. **How many milliliters of a 2000.0 mg/dL glucose stock solution are needed to prepare 100.0 mL of a 150.0 mg/dL glucose working standard?**

- A. 1.5 mL

- B. 7.5 mL
- C. 15.0 mL
- D. 25.0 mL

Laboratory Operations/Calculate/Solutions/2

Answers to Questions 4–7

4. **C** The MW of HNO_3 is 63.0 g. Because the valance of the acid is 1 (1 mole of hydrogen is produced per mole of acid), the equivalent weight is also 63.0 g. The mass needed is calculated by multiplying the normality (2.0 N) by the equivalent weight (63.0 g) and volume (1.0 L); therefore, 126.0 g of acid are required. Because the purity is 68.0% and the specific gravity 1.42, the amount of HNO_3 in grams per milliliter is 0.68×1.42 g/mL or 0.9656 g/mL. The volume required is calculated by dividing the mass needed (126.0 g) by the grams per milliliter.

$$\text{mL HNO}_3 = 126.0 \text{ g} \div 0.9656 \text{ g/mL} = 126.0 \text{ g} \times 1.0 \text{ mL}/0.9656 \text{ g} = 130.5 \text{ mL}$$

5. **C** The SI unit is the recommended method of reporting clinical laboratory results. The SI unit for all electrolytes is millimoles per liter (mmol/L). To convert from milligrams per deciliter to mmol/L, multiply by 10 to convert to milligrams per liter, then divide by the atomic mass expressed in milligrams.

$$10.0 \text{ mg/dL} \times 10.0 \text{ dL}/1.0 \text{ L} = 100.0 \text{ mg/L}$$

$$100.0 \text{ mg/L} \times 1.0 \text{ mmol}/40.1 \text{ mg} = 2.5 \text{ mmol/L}$$

6. **C** To convert from mEq/L to mg/dL, first calculate the milliequivalent weight (equivalent weight expressed in milligrams), which is the atomic mass divided by the valence. Because magnesium is divalent, each mole has the charge equivalent of 2 moles of hydrogen. Then, multiply the mEq/L by the milliequivalent weight to convert to milligrams per liter (mg/L). Next, divide by 10 to convert mg/L to mg/dL.

$$\text{Milliequivalent weight Mg} = 24.3 \div 2$$

$$= 12.15 \text{ mg/mEq}$$

$$2.0 \text{ mEq/L} \times 12.15 \text{ mg/mEq} = 24.3 \text{ mg/L}$$

$$24.3 \text{ mg/L} \times 1.0 \text{ L}/10.0 \text{ dL} = 2.4 \text{ mg/dL}$$

7. **B** To calculate the volume of stock solution needed, divide the concentration of working standard by the concentration of stock standard, and then multiply by the volume of working standard that is needed.

$$C_1 \times V_1 = C_2 \times V_2,$$

where C_1 = concentration of stock standard

V_1 = volume of stock standard

C_2 = concentration of working standard

V_2 = volume of working standard

$$2000.0 \text{ mg/dL} \times V_1 = 150.0 \text{ mg/dL} \times 100.0 \text{ mL}$$

$$V_1 = (150.0 \div 2000.0) \times 100.0 \text{ mL}$$

$$V_1 = 7.5 \text{ mL}$$

8. What is the pH of a solution of HNO_3 , if the hydrogen ion (H^+) concentration is $2.5 \times 10^{-2} \text{ M}$?

- A. 1.0
- B. 1.6
- C. 2.5
- D. 2.8

Laboratory Operations/Calculate/pH/2

9. Calculate the pH of a solution of 1.5×10^{-5} molar (M) ammonium hydroxide (NH_4OH).

- A. 4.2
- B. 7.2
- C. 9.2
- D. 11.2

Laboratory Operations/Calculate/pH/2

10. How many significant figures should be reported when the pH of a 0.060 M solution of HNO_3 is calculated?

- A. 1
- B. 2
- C. 3
- D. 4

Laboratory Operations/Calculate/Significant figures/2

11. What is the pH of a 0.05 M solution of acetic acid?

Dissociation constant (K_a) = 1.75×10^{-5} , $\text{p}K_a = 4.76$

- A. 1.7
- B. 3.0
- C. 4.3
- D. 4.6

Laboratory Operations/Calculate/pH/2

12. What is the pH of a buffer containing 40.0 mmol/L NaHC_2O_4 and 4.0 mmol/L $\text{H}_2\text{C}_2\text{O}_4$?

$\text{p}K_a = 1.25$

- A. 1.35
- B. 2.25
- C. 5.75
- D. 6.12

Laboratory Operations/Calculate/pH/2

Answers to Questions 8–12

8. **B** For a strong acid, the pH is equal to the negative logarithm of the H^+ concentration.

$$\text{pH} = -\text{Log } \text{H}^+$$

$$\text{pH} = -\text{Log } 0.025$$

$$\text{pH} = 1.6$$

9. **C** First, calculate the pOH of the solution.

$$\text{pOH} = -\text{Log } [\text{OH}^-]$$

$$pOH = -\text{Log } 1.5 \times 10^{-5} = 4.82$$

$$pH = 14 - pOH$$

$$pH = 14 - 4.8 = 9.2$$

- 10. B** When zeros appear by themselves to the left of the decimal point, they are not significant. When they are to the left of the decimal point and are preceded or followed by a number, they are significant. Zeros after the decimal point preceding a number are not significant. However, they are significant if they follow another number (either before or after the decimal point) or are between two numbers. Therefore, 0.060 M has only two significant figures (the underlined digits). In laboratory practice, most analytes are reported with two significant figures. Routine analytes that are exceptions are pH, which includes three significant figures, and analytes with whole numbers above 100, such as sodium, cholesterol, triglycerides, and glucose.

- 11. B** Weak acids are not completely ionized, and pH must be calculated from the dissociation constant of the acid (in this case 1.75×10^{-5}).

$$K_a = \frac{[H^+] \times [Ac^-]}{[HAc]}$$

$$1.75 \times 10^{-5} = \frac{[H^+] \times [Ac^-]}{5.0 \times 10^{-2}}$$

$$\text{Since } [H^+] = [Ac^-]$$

$$X^2 = (1.75 \times 10^{-5}) \times (5.0 \times 10^{-2}) = 8.75 \times 10^{-7}$$

$$x = \sqrt{8.75 \times 10^{-7}} = [H^+] = 9.35 \times 10^{-4} M$$

$$pH = -\text{Log } 9.35 \times 10^{-4} M = 3.0$$

$$\text{Alternatively, } pH = \frac{1}{2} (pK_a - \text{Log HA})$$

$$pH = \frac{1}{2} (4.76 - \text{Log } 5.0 \times 10^{-2})$$

$$= \frac{1}{2} (4.76 + 1.30)$$

$$= 3.0$$

- 12. B** The Henderson-Hasselbalch equation can be used to determine the pH of a buffer containing a weak acid and a salt of the acid.

$$pH = pK_a + \frac{\log \text{salt}}{\text{acid}}$$

$$= 1.25 + \log \frac{40.0 \text{ mmol/L}}{4.0 \text{ mmol/L}}$$

$$= 1.25 + \log 10$$

$$= 2.25$$

- 13. A** solvent needed for high-performance liquid chromatography (HPLC) requires a 20.0 mmol/L phosphoric acid buffer, pH 3.50, made by mixing potassium dihydrogen phosphate (KH_2PO_4) and phosphoric acid (H_3PO_4). How many grams of KH_2PO_4 are required to make 1.0 L of this buffer?

Formula weights: $\text{KH}_2\text{PO}_4 = 136.1$; $\text{H}_3\text{PO}_4 = 98.0$; $pK_a \text{ H}_3\text{PO}_4 = 2.12$

A. 1.96 g

- B. 2.61 g
- C. 2.72 g
- D. 19.2 g

Laboratory Operations/Calculate/Buffers/2

14. A procedure for cholesterol is calibrated with a serum-based cholesterol standard that was determined by the Abell-Kendall method to be 200.0 mg/dL. Assuming the same volume of sample and reagent are used, calculate the cholesterol concentration in the patient's sample from the following results:

Standard Concentration	Absorbance of Reagent Blank	Absorbance of Standard	Absorbance of Patient Serum
200 mg/dL	0.00	0.860	0.740

- A. 123 mg/dL
- B. 172 mg/dL
- C. 232 mg/dL
- D. 314 mg/dL

Laboratory Operations/Calculate/Beer's law/2

15. A glycerol kinase method for triglyceride calls for a serum blank in which lipase is absent from the reagent in order to measure endogenous glycerol. Given the following results, and assuming the same volume of sample and reagent are used for each test, calculate the triglyceride concentration in the patient's sample.

- A. 119 mg/dL
- B. 131 mg/dL
- C. 156 mg/dL
- D. 180 mg/dL

Laboratory Operations/Calculate/Beer's law/2

Standard Concentration	Absorbance of Reagent Blank	Absorbance of Standard	Absorbance of Patient Serum	Absorbance of Serum Blank
200 mg/dL	0.000	0.62	0.750	0.100

Answers to Questions 13–15

13. **B** The Henderson-Hasselbalch equation is used to calculate the ratio of salt to acid needed to give a pH of 3.50.

$$\begin{aligned}
 pH &= pK_a + \log(\text{salt}/\text{acid}) \\
 3.50 &= 2.12 + \log(KH_2PO_4/H_3PO_4) \\
 1.38 &= \log(KH_2PO_4/H_3PO_4) \\
 \text{Antilog } 1.38 &= KH_2PO_4/H_3PO_4 \\
 23.99 &= KH_2PO_4/H_3PO_4
 \end{aligned}$$

Rearranging gives $KH_2PO_4 = 23.99 \times H_3PO_4$. Because the phosphate in the buffer is 20.0 mmol/L, then $H_3PO_4 + KH_2PO_4$ must equal 20. Because $KH_2PO_4 = 23.99 \times H_3PO_4$,

then:

$$\begin{aligned}H_3PO_4 + (23.99 \times H_3PO_4) &= 20.0 \text{ mmol/L} \\24.99 \times H_3PO_4 &= 20.0 \text{ mmol/L} \\H_3PO_4 &= 20.0/24.99 = 0.800 \text{ mmol/L} \\KH_2PO_4 &= 20.0 - 0.800 = 19.2 \text{ mmol/L (0.0192 M)}\end{aligned}$$

To determine the grams required, multiply the moles of KH_2PO_4 by the formula weight.
 $0.0192 \text{ mol/L} \times 136.1 \text{ g/mol} = 2.613 \text{ g}$

14. **B** $C_u = A_u/A_s \times C_s$ where C_u = concentration of unknown, A_u = absorbance of unknown, A_s = absorbance of standard, and C_s = concentration of standard.

$$\begin{aligned}C_u &= 0.740/0.860 \times 200 \text{ mg/dL} \\&= 172 \text{ mg/dL}\end{aligned}$$

15. **B** The serum blank absorbance is subtracted from the result for the patient's serum before applying the ratiometric formula to calculate concentration.

$$\begin{aligned}C_u &= [(A_u - A_{SB})/A_s] \times C_s \\&\text{where} \\A_{SB} &= \text{absorbance of serum blank} \\&= (0.750 - 0.100)/0.620 \times 125 \text{ mg/dL} \\&= 131 \text{ mg/dL}\end{aligned}$$

16. **A** A procedure for aspartate aminotransferase (AST) is performed manually because of a repeating error code for nonlinearity obtained on the laboratory's automated chemistry analyzer; 0.05 mL of serum and 1.0 mL of substrate are used. The reaction rate is measured at 30°C at 340 nm using a 1.0 cm light path, and the delta absorbance ($-\Delta A$) per minute is determined to be 0.382. Based on a molar absorptivity coefficient for nicotinamide adenine dinucleotide (NADH) at 340 nm of $6.22 \times 10^3 \text{ M}^{-1} \text{ cm}^{-1} \text{ L}^{-1}$, calculate the enzyme activity in international units (IUs) per liter.

- A. 26 IU/L
- B. 326 IU/L
- C. 1228 IU/L
- D. 1290 IU/L

Laboratory Operations/Calculate/International units/2

17. **When referring to conventional quality control (QC) results, what parameter usually determines the acceptable range?**

- A. The 95% confidence interval for the mean
- B. The range that includes 50% of the results
- C. The central 68% of results
- D. The range encompassed by ± 2.5 standard deviations(s)

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Quality control/1

18. **Which of the following QC rules would be broken one out of 20 times by chance alone?**

- A. 1_{2s}

- B. 2_{2s}
- C. 1_{3s}
- D. 1_{4s}

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Quality control/1

19. Using conventional QC rules, which of the following conditions is cause for rejecting an analytical run?

- A. Two consecutive controls greater than 2s above or below the mean
- B. Three consecutive controls greater than 1s above the mean
- C. Four controls steadily increasing in value but less than $\pm 1s$ from the mean
- D. One control above +1s and the other below -1s from the mean

Laboratory Operations/Select course of action/Quality control/3

Answers to Questions 16–19

- 16. D** An international unit (IU) is defined as 1 μmol (micromole) of substrate consumed or product produced per minute. The micromoles of NADH consumed in this reaction are determined by dividing the change in absorbance per minute by the absorbance of 1 μmol of NADH. Because 1 mol/L/cm would have an absorbance of 6.22×10^3 absorbance units, then 1 $\mu\text{mol/mL/cm}$ would produce an absorbance of 6.22. Therefore, dividing the ΔA per minute by 6.22 gives the micromoles of NADH consumed in the reaction. This is multiplied by the dilution of serum to determine the micromoles per minute per milliliter, and multiplied by 1,000 to convert to liters. where TV is total volume and SV is sample volume

$$\text{IU/L} = \frac{\Delta A/\text{min} \times \text{TV}(\text{mL}) \times 1,000 \text{ mL/L}}{6.22(A/\mu\text{mol/mL/cm}) \times 1 \text{ cm} \times \text{SV}(\text{mL})}$$

where TV is total volume and SV is sample volume

$$= \frac{\Delta A/\text{min} \times 1.05 \times 1,000}{6.22 \times 0.05}$$

$$= \Delta A/\text{min} \times \frac{1,050}{0.311}$$

$$= \Delta A/\text{min} \times 3,376$$

$$= 0.382 \times 3376 = 1,290 \text{ IU/L}$$

- 17. A** The acceptable range for QC results is often set at the 95% confidence interval. This is defined as the range between $-1.96s$ and $+1.96s$. This means that we can expect a QC result to fall within this range 95 out of 100 times. For practical purposes, this is the same as $\pm 2s$ (95.4 out of 100 results should fall within $\pm 2s$ of the mean on the basis of chance). However, this limit will result in a high rate of false rejections (rejecting good results) when the assay has a low bias and coefficient of variation (CV) (in other words, performs significantly better than the allowable error limit for the test).

- 18. A** The notation 1_{2s} means that one control is outside ± 2 standard deviation units. QC

results follow the bell-shaped curve called the *Gaussian (normal) distribution*. If a control is assayed 100 times, 68 out of 100 results would fall within $+1s$ and $-1s$ of the mean. Ninety-five (95.4) out of 100 results would fall within $+2s$ and $-2s$. This leaves only 5 out of 100 results (1:20) that fall outside the $\pm 2s$ limit. Also, 99.7 out of 100 results fall within $\pm 3s$ of the mean.

- 19. A** Rejecting a run when three consecutive controls fall between 1 and 2s or when a trend of four increasing or decreasing control results occurs would lead to frequent rejection of valid analytical runs. Appropriate control limits are four consecutive controls above or below 1s (4_{1s}) to detect a significant shift, and a cumulative sum control chart (CUSUM) result exceeding the $\pm 2.7s$ limit to detect a significant shift or trend. When controls deviate in opposite directions, the difference should exceed 4s before the run is rejected.

- 20. One of two controls within a run is above $+2s$ and the other control is below $-2s$ from the mean. What do these results indicate?**

A. Poor precision has led to random error (RE)
B. A systematic error (SE) is present
C. Proportional error is present
D. QC material is contaminated

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/2

- 21. Two consecutive controls are both beyond $-2s$ from the mean. How frequently would this occur on the basis of chance alone?**

A. 1:100
B. 5:100
C. 1:400
D. 1:1,600

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Quality control/2

- 22. The term R_{4s} means that:**

A. Four consecutive controls are greater than ± 1 standard deviation from the mean
B. Two controls in the same run are greater than 4s units apart
C. Two consecutive controls in the same run are each greater than $\pm 4s$ from the mean
D. There is a shift above the mean for four consecutive controls

Chemistry/Evaluate laboratory data to assess validity/Accuracy of procedures/Quality control/2

- 23. A trend in QC results is most likely caused by:**

A. Deterioration of the reagent
B. Miscalibration of the instrument
C. Improper dilution of standards
D. Electronic noise

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Quality control/2

- 24. In most circumstances, when two controls within a run are both greater than $\pm 2s$ from the mean, what action should be taken first?**

A. Recalibrate, and then repeat controls followed by selected patient samples if QC is

acceptable

- B. Repeat the controls before taking any corrective action
- C. Change the reagent lot, and then recalibrate
- D. Prepare fresh standards, and recalibrate

Laboratory Operations/Evaluate laboratory data to take corrective action according to predetermined criteria/Quality control/3

Answers to Questions 20–24

20. **A** When control results deviate from the mean in opposite direction, the run is affected by RE, which results from imprecision. An analytical run is rejected when two controls within the same run have an algebraic difference in excess of $4s$ (R_{4s}). The R_{4s} rule is applied only to controls within a run (level 1 – level 2), never across runs or days.
21. **D** QC results follow a Gaussian or normal distribution. Ninety-five percent of the results fall within $\pm 2s$ of the mean; therefore, 2.5 out of 100 (1:40) are above $+2s$ and 2.5 out of 100 are below $-2s$. The probability of two consecutive controls being beyond $-2s$ is the product of their individual probabilities. $1/40 \times 1/40 = 1/1,600$ trials by chance.
22. **B** The R_{4s} rule is applied to two control levels within the same run. The rule is violated when the algebraic difference between them (level 1 – level 2) exceeds $4s$. The rule is never applied across different runs. The R_{4s} rule detects RE (error caused by lack of precision).
23. **A** A trend occurs when six or more consecutive QC results either increase or decrease in the same direction; however, this is not cause for rejection until a multirule is broken. Trends are SEs (affecting accuracy) linked to an unstable reagent, calibrator, or instrument condition. For example, loss of volatile acid from a reagent causes a steady pH increase, preventing separation of analyte from protein. This results in lower QC results each day.
24. **A** When the 2_{2s} rule is broken, this usually means that excessive SE is present and corrective action is required (repeating just the QC will not correct the problem). If recalibration yields acceptable QC results, both sets of QC results and the corrective action taken are documented in the QC log. If the magnitude of the error is large enough to be medically significant, then all patient samples since the last previously acceptable QC should be repeated. If in question, the magnitude of the error can be evaluated by repeating abnormal patient samples. If the average difference between results before and after recalibration is greater than $2s$ or error limit, then all samples should be repeated since the last acceptable QC.

25. When establishing a QC acceptable range, which of the following practices is inappropriate?

- A. Using last month's QC data to determine current target limits
- B. Exclusion of any QC results greater than $\pm 2s$ from the mean
- C. Using control results from all shifts on which the assay is performed
- D. Using limits determined by reference laboratories using the same method

Laboratory Operations/Apply principles of laboratory operations/Quality control/2

26. Which of the following assays has the poorest precision?

	Analyte	Mean (mmol/L)	Standard Deviation
A.	Calcium (Ca)	2.5	0.3
B.	Potassium (K)	4.0	0.4
C.	Sodium (Na)	140	4.0
D.	Chloride (Cl)	100	2.5

Laboratory Operations/Calculate/Coefficient of variation/3

27. Given the following data, calculate the coefficient of variation for glucose:

Analyte	Mean	Standard Deviation
Glucose	76 mg/dL	2.3

- A. 3.0%
- B. 4.6%
- C. 7.6%
- D. 33.0%

Laboratory Operations/Calculate/Statistics/2

28. Which of the following plots is best for detecting all types of QC errors?

- A. Levy-Jennings
- B. Tonks-Youden
- C. CUSUM
- D. Linear regression

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/2

29. Which of the following plots is best for comparison of precision and accuracy among laboratories?

- A. Levy-Jennings
- B. Tonks-Youden
- C. CUSUM
- D. Linear regression

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/2

Answers to Questions 25–29

25. **B** Data between $\pm 2s$ and $\pm 3s$ must be included in calculations of the next month's acceptable range. Elimination of these values would continuously reduce the distribution of QC results, making "out of control" situations a frequent occurrence. Generally, QC results greater than $3s$ are not used to calculate next month's mean.
26. **A** Although calcium has the lowest s , it represents the assay with poorest precision. Relative precision between different analytes or different levels of the same analyte must be evaluated by the CV because standard deviation is dependent on sample concentrations. $CV = (s \times 100)/\text{Mean}$. This normalizes standard deviation to a mean of 100. The CV for calcium in the example is 12.0%.

27. **A** The coefficient of variation is calculated by dividing the standard deviation by the mean and multiplying by 100.

$$\begin{aligned}\%CV &= \frac{s}{\bar{X}} \times 100 \\ &= \frac{2.3}{76} \times 100 = 3.0\%\end{aligned}$$

CV is the most appropriate statistic to use when comparing the precision of samples that have different means. For example, when comparing the precision of the level 1 control to the level 2 control, the coefficient of variation normalizes the variance to be independent of the mean. The analysis is more precise for the control with the lower CV.

28. **A** The Levy-Jennings plot is a graph of all QC results with concentration plotted on the y-axis and run number on the x-axis. The mean is at the center of the y-axis, and concentrations corresponding to -2 and $+2s$ (or other specified control limits) are highlighted. Results are evaluated for multirule violations across both levels and runs. Corrective action for shifts and trends can be taken before QC rules are broken.
29. **B** The Tonks-Youden plot is used for interlaboratory comparison of monthly means. The method mean for level 1 is at the center of the y-axis and mean for level 2 at the center of the x-axis. Lines are drawn from the means of both levels across the graph, dividing it into four equal quadrants. If a laboratory's monthly means both plot in the lower left or upper right quadrants, then SE exists in its method.

30. Which plot will give the earliest indication of a shift or trend?

- A.** Levy-Jennings
- B.** Tonks-Youden
- C.** CUSUM
- D.** Histogram

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/2

31. All of the following are requirements for a QC material except:

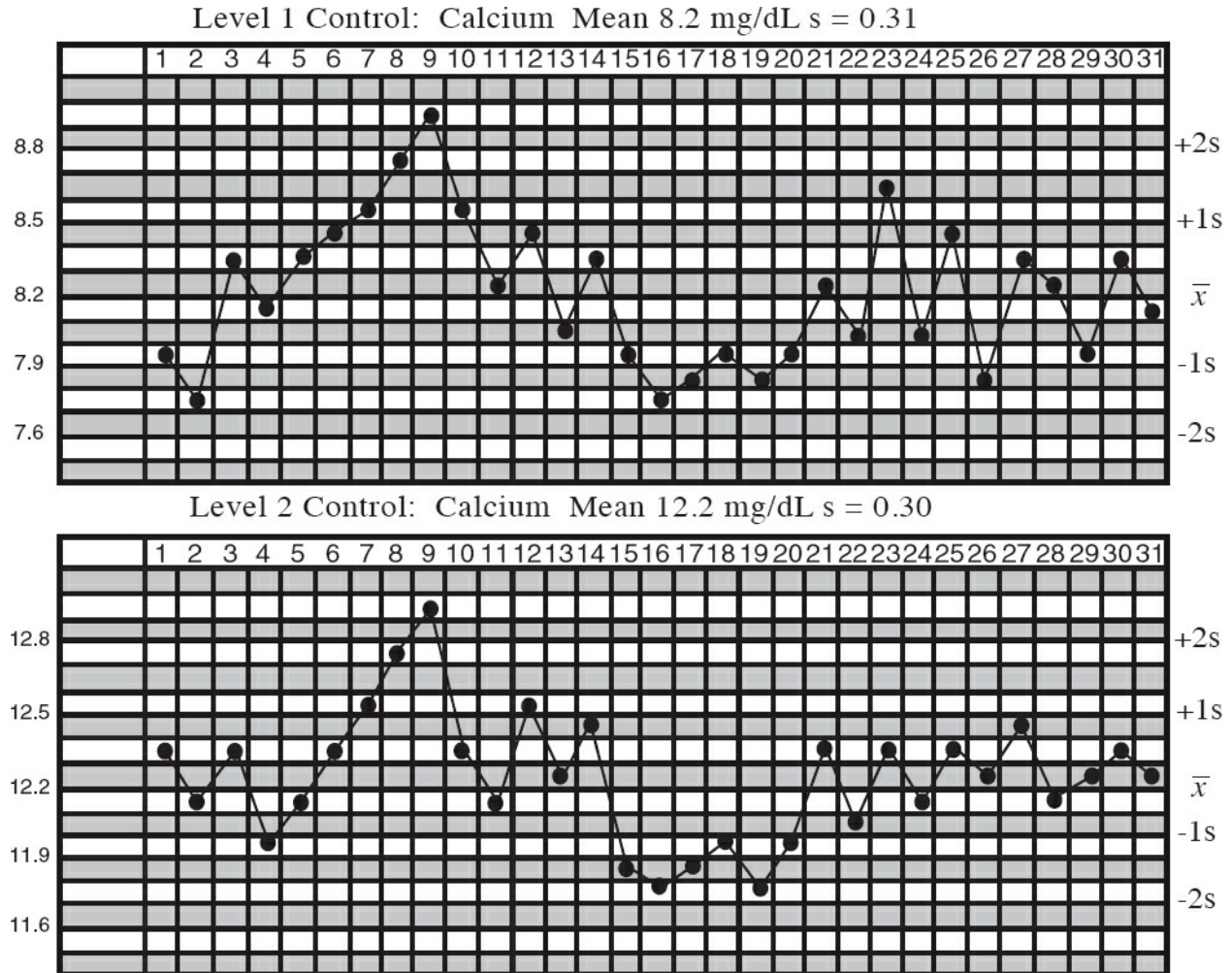
- A.** Long-term stability
- B.** The matrix is similar to the specimens being tested
- C.** The concentration of analytes reflects the clinical range
- D.** Analyte concentration must be independent of the method of assay

Laboratory Operations/Apply principles of basic laboratory procedures/Quality control/2

Answers to Questions 30–31

30. **C** CUSUM points are the algebraic sum of the difference between each QC result and the mean. The y-axis is the sum of differences and the x-axis is the run number. The center of the y-axis is 0. Because QC results follow a random distribution, the points should distribute about the zero line. Results are out of control when the slope exceeds 45 degrees or a decision limit (e.g., $\pm 2.7s$) is exceeded.
31. **D** Quality control materials are stable, made of the same components as the specimen,

cover the dynamic linear range of the assay, and can be used for multiple analytes. The target mean for QC samples is determined from replicate assays by the user's method, not the "true" concentration of the analyte. Out-of-control results are linked to analytic performance rather than to the inherent accuracy of the method.



Questions 32–35 refer to the preceding Levy–Jennings chart.

32. Examine the Levy-Jennings chart at the bottom of the previous page, and identify the QC problem that occurred during the first half of the month.

- A. Shift
- B. Trend
- C. Random error
- D. Kurtosis

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/3

33. Referring to the Levy-Jennings chart, what is the first day in the month when the run should be rejected and patient results should be repeated using conventional QC limits and multirules?

- A. Day 6
- B. Day 7

- C. Day 8
- D. Day 9

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/3

34. Referring to the Levy-Jennings chart, what analytical error is present during the second half of the month?

- A. Shift
- B. Trend
- C. Random error
- D. Kurtosis

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/3

35. What is the first day in the second half of the month that patient results would be rejected using conventional QC limits and multirules?

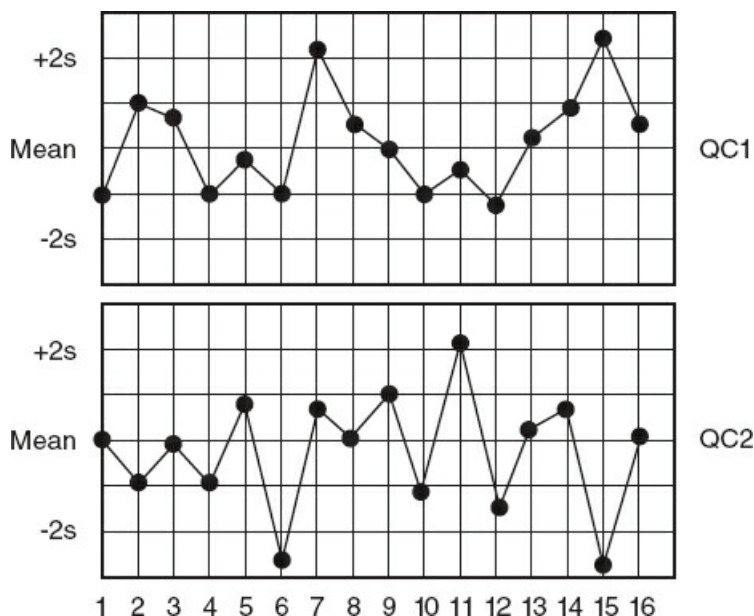
- A. Day 16
- B. Day 17
- C. Day 18
- D. Day 19

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/3

Answers to Questions 32–35

- 32. B** A trend is characterized by six consecutive decreasing or increasing control results. The value for both controls becomes progressively higher from day 4 to day 9. Trends are caused by changes to the test system that increase over time, such as deterioration of reagents or calibrators, progressive changes in temperature, evaporation, light exposure, and bacterial contamination. A trend is a type of SE because all results are affected. Conversely, RE affects some results (but not others) in an unpredictable manner. Control rules affected by RE are 1_{3s} and R_{4s} .
- 33. C** Although the trend is apparent across QC levels by day 7, the patient results would not be rejected until day 8 when the 4_{1s} rule is broken. An advantage to plotting control data is that trends can be identified before results are out of control and patient data must be rejected. In this case, corrective steps should have been implemented by day 7 to avoid the delay and expense associated with having to repeat the analysis of patient samples. Because no action was taken on day 7 or 8, the 2_{2s} rule was broken on day 9.
- 34. A** A shift is characterized by six consecutive points lying on the same side of the mean. This occurs from day 15 to day 20. Shifts are caused by a change in the assay conditions that affect the accuracy of all results, such as a change in the concentration of the calibrator; change in reagent; a new lot of reagent that differs in composition; or improper temperature setting, wavelength, or sample volume. The term *kurtosis* refers to the degree of flatness or sharpness in the peak of a set of values having a Gaussian distribution.
- 35. B** The 4_{1s} rule is broken across QC levels on day 17. This means that four consecutive controls are greater than $\pm 1s$ from the mean. QC rules that are sensitive to SE are applied across both runs and levels to increase the probability of error detection. These are 2_{2s} , 4_{1s} , and $10_{\bar{x}}$.

36. Given the following QC chart, identify the day in which a violation of the R_{4s} QC rule occurs:



- A. Day 3
- B. Day 8
- C. Day 10
- D. Day 15

Laboratory Operations/Evaluate laboratory data to recognize problems/Quality control/3

37. What is the minimum requirement for performing QC for a total protein assay?

- A. One level assayed every 8 hours
- B. Two levels assayed within 8 hours
- C. Two levels assayed within 24 hours
- D. Three levels assayed within 24 hours

Laboratory Operations/Apply principles of basic laboratory procedures/Quality control/2

38. Which of the following statistical tests is used to compare the means of two methods?

- A. Student's t test
- B. F distribution
- C. Correlation coefficient (r)
- D. Linear regression analysis

Laboratory Operations/Evaluate laboratory data to assess the validity/Accuracy of procedures/Statistics/2

Answers to Questions 36–38

36. **D** An R_{4s} error is defined as the algebraic difference between two controls within the same run. In this Levy-Jennings plot, on day 15, Level 1 is above the $+2s$ limit (approximately $+2.5s$) and Level 2 is below the $-2s$ limit (approximately $-2.5s$). These controls are approximately $5s$ apart ($+2.5s$ minus $-2.5s = +5s$).

37. C The minimum requirement by federal regulations (subpart K of the Clinical Laboratory Improvement Act, 1988 [CLIA 1988]) for frequency of QC for a general chemistry analyte is two levels of control assayed every 24 hours, provided the manufacturer's QC requirements are no greater. Some laboratories prefer to assay two control levels every 8 hours to increase the opportunity for error detection. Two controls every 8 hours are required for blood gases, automated hematology, and point-of-care glucose testing to comply with College of American Pathology requirements. Three levels of control every 24 hours should be used for analytes that have significantly different CVs at low, normal, and high concentrations. These include blood gases, therapeutic drugs, and hormones.

38. A Student's t test is the ratio of mean difference to the standard error of the mean difference (bias/RE) and tests for a significant difference in means. The F test is the ratio of variances and determines if one method is significantly less precise. The correlation coefficient is a measure of the association between two variables and should be high in any method comparison. An r value less than 0.90 in method comparisons usually occurs when the range of results is too narrow.

39. Two freezing point osmometers are compared by running 40 paired patient samples one time on each instrument, and the following results are obtained:

Instrument	Mean	Standard Deviation
Osmometer A	280 milliosmoles per kilogram(mOsm/kg)	3.1
Osmometer B	294 mOsm/kg	2.8

If the critical value for $F = 2.8$, then what conclusion can be drawn regarding the precision of the two instruments?

- A.** There is no statistically significant difference in precision
- B.** Osmometer A demonstrates better precision that is statistically significant
- C.** Osmometer B demonstrates better precision that is statistically significant
- D.** Precision cannot be evaluated statistically when single measurements are made on samples

Laboratory Operations/Evaluate laboratory data to assess the validity/Accuracy of procedures/Statistics/3

40. Two methods for total cholesterol are compared by running 40 paired patient samples in duplicate on each instrument. The following results are obtained:

Instrument	Mean	Standard Deviation
Method x (reference method)	235 mg/dL	3.8
Method y (candidate method)	246 mg/dL	3.4

Assuming the samples are collected and stored in the same way and the analysis done by a medical laboratory scientist (MLS) who is familiar with both methods, what is the bias of method y?

- A.** 0.4
- B.** 7.2

C. 10.6

D. 11.0

Laboratory Operations/Evaluate laboratory data to assess the validity/Accuracy of procedures/Statistics/2

41. When the magnitude of error increases with increasing sample concentration, it is called:

A. Constant error

B. Proportional error

C. Random error

D. Bias

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Statistics/2

Answers to Questions 39–41

39. A The F test determines whether there is a statistically significant difference in the variance of the two sampling distributions. Assuming the samples are collected and stored in the same way and the analysis is done by a person who is familiar with the instruments, then differences in variance can be attributed to a difference in instrument precision. The F test is calculated by dividing the variance $(s_1)^2$ of the instrument having the higher standard deviation by the variance $(s_2)^2$ of the instrument having the smaller standard deviation.

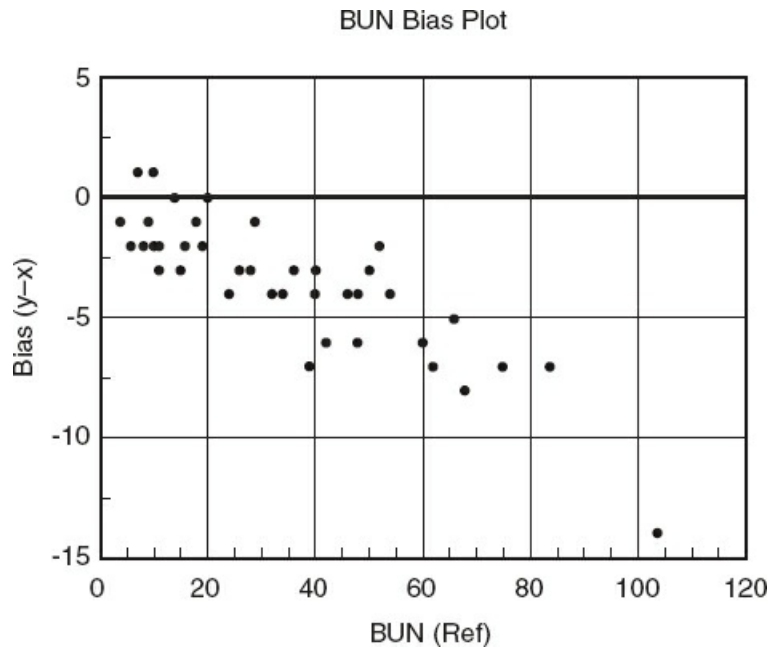
$$F = (s_1)^2 \div (s_2)^2 = (3.1)^2 \div (2.8)^2 = 9.61 \div 7.84 = 1.22$$

If the value of F is smaller than the critical value at the 0.10 level of significance, then the hypothesis (there is no significant difference in the variance of the two instruments) is accepted. The 0.10 significance level is used for one-tailed statistical tests.

40. D Bias is defined as the difference between the means of the two methods and is calculated by using the formula: $\text{bias} = \bar{y} - \bar{x}$. The bias is an estimate of SE. The student's t test is used to determine if bias is statistically significant. The t statistic is the ratio of bias to the standard error of the mean difference. The greater the bias, the higher the t score.

41. B Proportional error (slope or percent error) results in greater absolute error (deviation from the target value) at higher sample concentration. Constant error refers to a difference between the target value and the result, which is independent of sample concentration. For example, if both level 1 and level 2 controls for laboratory A average 5 mg/dL below the cumulative mean reported by all other laboratories using the same method, then laboratory A has a constant error of –5 mg/dL for that method. Both proportional and constant error contribute to method bias.

42. Which explanation is the best interpretation of the following blood urea nitrogen (BUN) bias plot?



- A. The new method consistently overestimates BUN by a constant concentration
 - B. The new method is greater than the reference method but not by a statistically significant margin
 - C. The new method is lower than the reference method by 5 mg/dL
 - D. The new method is lower than the reference and the magnitude is concentration dependent
- Laboratory Operations/Evaluate laboratory data to assess the validity/Accuracy of procedures/Statistics/3*

43. Serum samples collected from hospitalized patients over a 2-week period were split into two aliquots and analyzed for prostate specific antigen (PSA) by using two methods. Each sample was assayed within 30 minutes of collection by a scientist familiar with both methods. The reference method was method x (upper reference limit = 4.0 µg/L). Linear regression analysis was performed by the least-squares method, and results are as follows:

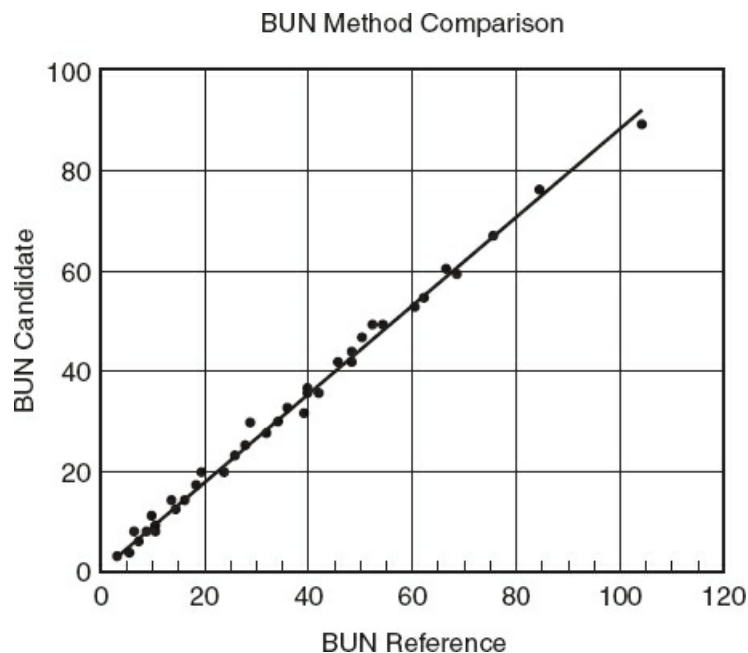
Linear Regression	Correlation Coefficient (r)	Standard Error of Estimate ($s_{y/x}$)
$\hat{y} = 2.10 + 1.01x$	0.984	0.23

Which statement best characterizes the relationship between the methods?

- A. There is a significant bias caused by constant error
 - B. There is a significant proportional error
 - C. There is no disagreement between the methods because the correlation coefficient approaches 1.0
 - D. There is no SE, but the RE of the new method is unacceptable
- Laboratory Operations/Evaluate laboratory data to assess the validity/Accuracy of procedures/Statistics/2*

42. **D** A bias (Bland-Altman) plot compares bias (candidate method – reference method) to the result of the reference method. Ideally, points should be scattered equally on both sides of the zero line. When the majority of points are below the zero line, the candidate method is negatively biased (lower than the reference). In this case, the difference between the methods increases in proportion to the BUN concentration. This type of scatter occurs when the slope of the linear regression line is low.
43. **A** The linear regression analysis is the most useful statistic to compare paired patient results because it estimates the magnitude of specific errors. The y-intercept of the regression line is a measure of constant error, and the slope is a measure of proportional error. Together, these represent the bias or SE of the new method. The correlation coefficient is influenced by the range of the sample and RE. Two methods that measure the same analyte will have a high correlation coefficient, provided the concentrations are measured over a wide range, and this statistic should not be used to judge the acceptability of the new method. The standard error of estimate is a measure of the closeness of data points to the regression line and is an expression of RE.

44. Which statement best summarizes the relationship between the new BUN method and reference method based on the following linear regression scatterplot?



- A. The methods agree very well but show a high standard error of estimate
- B. There is little or no constant error, but some proportional error
- C. There will be a significant degree of uncertainty in the regression equation
- D. There is significant constant and proportional error but little RE

Laboratory Operations/Evaluate laboratory data to assess the validity/Accuracy of procedures/Statistics/3

45. **A** new method for BUN is evaluated by comparing the results of 40 paired patient samples to the urease–UV method. Normal and high controls were run on each shift for 5 days, five times per day. The results are as follows:

Linear Regression	Low Control	High Control
$\hat{y} = -0.3 + 0.90x$	$\bar{x} = 14.2 \text{ mg/dL};$ $s = 1.24$	$\bar{x} = 48.6 \text{ mg/dL};$ $s = 1.12$

What is the total analytical error (TE) estimate for a sample having a concentration of 50 mg/dL?

- A. -2.2 mg/dL
- B. -2.8 mg/dL
- C. -7.5 mg/dL
- D. -10.0 mg/dL

Laboratory Operations/Calculate/Method comparison statistics/3

46. In addition to the number of true negatives (TNs), which of the following measurements is needed to calculate specificity?

- A. True positives (TPs)
- B. Prevalence
- C. False negatives (FNs)
- D. False positives (FPs)

Laboratory Operations/Calculation/Specificity/2

Answers to Questions 44–46

44. B The scatterplot shows that each sample produces a coordinate (x corresponds to the reference result and y to the candidate method result) that is very close to the regression line. This means that the variance of regression is low and there is a high degree of certainty that the predicted value of y will be close to its measured value. Near zero concentration there is good agreement between methods; however, the higher the result, the greater is the difference between x and y . The regression equation for this scatterplot is $\hat{y} = -0.01 + 0.90x$, indicating a proportional error of -10%.

45. C Linear regression analysis gives an estimate of SE, which is equal to $(\hat{y} - x_c)$, where x_c is the expected concentration, and $\hat{y}$ is the value predicted by the linear regression equation.

$$SE = [-0.3 + (0.9 \times 50 \text{ mg/dL})] - 50.0 \text{ mg/dL} = 44.7 - 50.0 = -5.3 \text{ mg/dL}$$

The standard deviation of the new method for the high control is used to estimate the RE because the mean of this control is nearest to the expected concentration of 50 mg/dL. RE is estimated by $\pm 1.96 \times s$.

$$RE = 1.96 \times 1.12 = \pm 2.2 \text{ mg/dL}$$

TE is equal to the sum of SE and RE.

$$TE = SE + RE = -5.3 \text{ mg/dL} + (-2.2 \text{ mg/dL}) = -7.5 \text{ mg/dL}$$

46. D The clinical specificity of a laboratory test is defined as TNs divided by the sum of TNs and FPs.

$$\% \text{ Specificity} = \frac{TN \times 100}{TN + FP}$$

Specificity is defined as the percentage of disease-free people who have a negative test result. The probability of FPs is calculated from the specificity as:

$$1 - \left(\frac{\% \text{ specificity}}{100} \right)$$

- 47. A new tumor marker for ovarian cancer is evaluated for sensitivity by testing serum samples from patients who have been diagnosed by staging biopsy specimens as having malignant or benign lesions. The following results were obtained:**

Number of malignant patients who are positive for CA 125 = 21 out of 24

Number of benign patients who are negative for CA 125 = 61 out of 62

What is the sensitivity of the new CA-125 test?

- A. 98.4%
- B. 95.3%
- C. 87.5%
- D. 85.0%

Clinical chemistry/Calculate/Sensitivity/2

- 48. A new test for prostate cancer is found to have a sensitivity of 80.0% and a specificity of 84.0%. If the prevalence of prostate cancer is 4.0% in men ages greater than 42 years, what is the predictive value of a positive test result (PVP) in this group?**

- A. 96.0%
- B. 86.0%
- C. 32.4%
- D. 17.2%

Chemistry/Calculate/Predictive value/2

- 49. What measurement in addition to TNs and prevalence is required to calculate the predictive value of a negative test result (PVN)?**

- A. FNs
- B. Variance
- C. TPs
- D. FPs

Chemistry/Calculate/Predictive value/2

- 50. A laboratory is establishing a reference range for a new analyte and wants the range to be determined by the regional population of adults ages 18 years or greater. The analyte concentration is known to be independent of race and gender. Which is the most appropriate process to follow?**

- A. Determine the mean and standard deviation of the analyte from 40 healthy adults, and calculate the $\pm 2s$ limit
- B. Measure the analyte in 120 healthy adults, and calculate the central 95th percentile
- C. Measure the analyte in 120 healthy adults, and use the lowest and highest as the reference range limits
- D. Measure the analyte in 60 healthy adults and 60 adults with conditions that affect the analyte concentration, and calculate the concentration of least overlap

Laboratory Operations/Select methods/Statistics/2

Answers to Questions 47–50

- 47. C** Sensitivity is defined as the percentage of persons with the disease who have a

positive test result. It is calculated as TPs (the number of people with the disease who test positive) divided by the sum of TPs and FNs.

$$\% \text{ Sensitivity} = \frac{TP \times 100}{TP + FN}$$

$$\text{Sensitivity} = (21 \times 100) \div (21 + 3) = 87.5\%$$

$$\text{Sensitivity} = (21 \times 100) \div (21 + 3) = 87.5\%$$

- 48. D** The PVP is defined as the percentage of persons with a positive test result who will have the disease or condition. It is dependent on the sensitivity of the test and the prevalence of the disease in the population tested. PVP is calculated by multiplying the TPs by 100 and then dividing by the sum of TPs and FPs.

$$\% \text{ PVP} = \frac{TP \times 100}{(TP + FP)}$$

where TP equals (sensitivity $\times$ prevalence) and FP equals $(1 - \text{specificity}) \times (1 - \text{prevalence})$

$$= \frac{0.80 \times 0.04 \times 100}{(0.80 \times 0.04) + [(1 - 0.84) \times (1 - 0.04)]}$$

$$= \frac{0.032 \times 100}{0.032 + (0.96 \times 0.16)}$$

$$= 17.2\%$$

- 49. A** PVN is defined as the probability that a person with a negative test result is free of disease. A high PVN is a characteristic of a good screening test. The predictive value of a negative test is calculated by multiplying the TNs by 100 and then dividing by the sum of the TNs and FNs.

$$\% \text{ PVN} = \frac{TN \times 100}{TN + FN}$$

- 50. B** Because the concentration of an analyte may not be normally distributed in a population, the reference range should not be determined from the standard deviation. It is more appropriate to determine the central 95th percentile (the range that encompasses 95% of the results). A minimum of 120 samples is needed for statistical significance. Results are rank ordered from lowest to highest. The third result is the lowest value and the 118th is the highest value in the reference range. The laboratory can verify a preexisting reference range (e.g., as determined by the manufacturer's study) by testing at least 20 healthy persons. If no more than 10% fall outside the range, it can be considered valid for the patient population.

- 51. When comparing the laboratory's monthly mean to its peer group to determine if bias is present, what statistic is most appropriate?**

- A. F test
- B. Linear regression analysis
- C. Correlation coefficient
- D. Standard deviation index (SDI)

Laboratory Operations/Apply principles of laboratory operations/Quality management/2

- 52. Which of the following methods is most useful to detect sample misidentification?**

- A. Cumulative summation

- B. Critical limit
- C. Delta limit
- D. Significant change limit

Laboratory Operations/Apply knowledge to identify sources of error/Statistics/2

53. Which of the following total quality management (TQM) tools can be used to calculate the analytical error rate for an analyte in the clinical laboratory?

- A. Lean production
- B. Six sigma
- C. International Standards Organization (ISO) 9000
- D. Laboratory information system

Laboratory Operations/Apply principles of laboratory operations/Quality management/2

Answers to Questions 51–53

- 51. D** The SDI equals the laboratory's mean minus the peer group's mean divided by the peer group's standard deviation. This normalizes the value and allows for performance comparisons for any analyte. It has a similar probability distribution to a t test and a value greater than 2.0 is considered significant.
- 52. C** Cumulative summation is a statistical method used in quality assurance to detect a trend in QC results. Critical limits are used to define when medical intervention is likely needed to prevent injury or death. The significant change limit is the difference in test results that is medically significant, or that which cannot be attributed to the sum of normal physiological and analytical variation. The delta limit (delta check) determines whether the difference between two measurements usually 24 to 48 hours apart exceeds the expected. Although this can result from an abrupt change in the patient's status, other causes are sample misidentification, contamination, and RE. Delta limits are expressed in percent and vary depending on analyte stability.
- 53. B** All four of the answer choices are TQM tools used in the clinical laboratory to improve performance. Six sigma is a measurement of the frequency of product defects. In clinical laboratories, it refers to the frequency of an erroneous result. At the six-sigma level, the analytical process has such small variance that an error of six times the standard deviation would still be within acceptable limits for total allowable error. For example, a six-sigma process for an analyte produces a significant error in test result only 3.4 out of 1 million times the test is performed. Conversely, a method performing at the three-sigma level would give 66,807 errors per million.

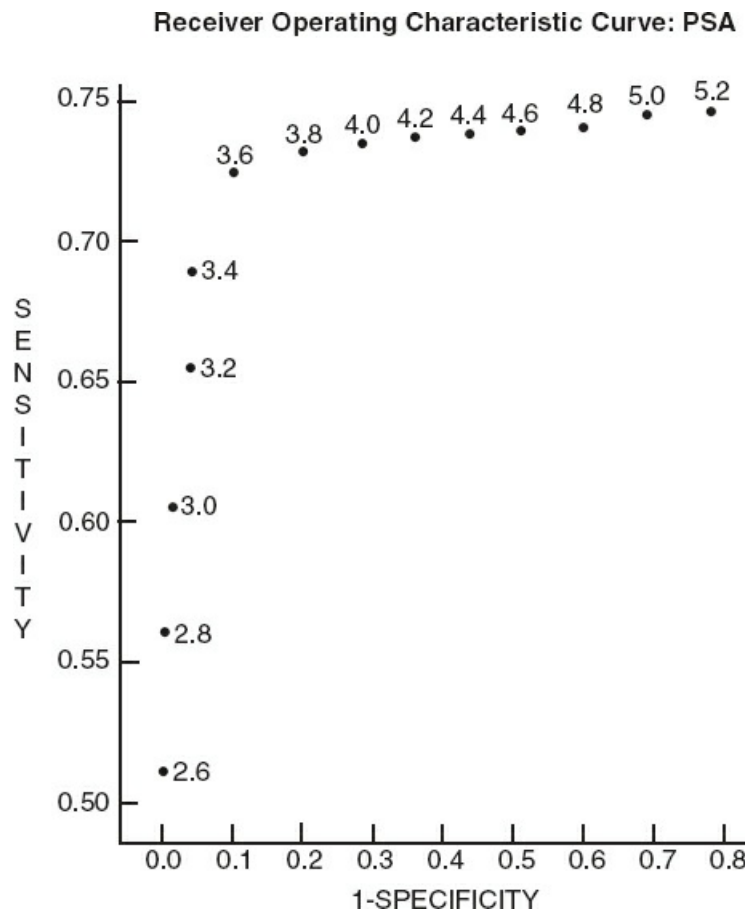
54. In which circumstances is a validation study (versus performing routine QC) required?

- A. Instrument recalibration
- B. Source lamp or ion selective electrode change
- C. Change in reagent lot
- D. Change in calibrator lot

Laboratory Operations/Apply principles of laboratory operations/Quality management/2

55. The following plot represents a study of a screening test for malignant prostate cancer using plasma PSA (nanograms per milliliter [ng/mL]). The outcome measured was positive cytology results obtained through biopsy analysis. What concentration gives the

highest sensitivity with the least number of unnecessary biopsy procedures?



- A. 2.6
- B. 3.6
- C. 3.8
- D. 5.2

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Laboratory operations/3

Answers to Questions 54–55

- 54. C** All of the listed conditions except a change in the reagent lot number can be validated by assaying two levels of control material after the procedure. A change in reagent lot number may alter the test system more dramatically, especially when the reagent was subjected to storage and shipping conditions that alter its performance. Therefore, both controls and patient samples should be analyzed and the results compared with the reagent in current use by utilizing criteria determined by the total allowable error for the analyte.
- 55. B** A receiver operating characteristic (ROC) curve is used to identify the test result, giving the highest sensitivity with the least number of FPs. Sensitivity (TPs) is plotted against FPs. The number in the uppermost left corner represents the highest detection

with the lowest number of FPs. In this case, a result of 3.6 ng/mL detects 72% of malignancies with one in 10 (10%) FPs.

56. Which statement below *best* describes the QC requirements for clinical laboratories?

- A. QC must be performed for all laboratory tests
- B. QC for nonwaived tests must include at least one level of control every 24 hours
- C. Equivalent quality control (eQC) is permitted for single device and point-of-care tests
- D. An individualized quality control plan (IQCP) can be used if approved by the lab's accrediting agency

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Quality control/1

57. What two parameters must be addressed in the risk assessment of an IQCP?

- A. Risk of testing error and severity of harm when test failure occurs
- B. Sigma level of the test and coefficient of variation index (CVI)
- C. Bias and uncertainty of the test result
- D. Analytical sensitivity and specificity

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Quality control/1

58. A laboratory's test for calcium has a bias of 0.5 mg/dL when compared with its peer group for proficiency testing. The CLIA total allowable error for serum or plasma calcium is ± 1.0 mg/dL. The laboratory's standard deviation for this test is 0.25 mg/dL. What is the laboratory's sigma?

- A. 50
- B. 2
- C. 1
- D. 0.25

Lab Operations/Calculate/Sigma/2

59. Which best describes the appropriate way to use QC decision limits for a test?

- A. Always use the 2_{2s} rule with or without others to determine rejection
- B. If QC is performed twice per day, use a combination of 2_{2s} , 1_{3s} , and 4_{1s}
- C. If QC is performed more than twice per day, use only the 2_{2s} rule
- D. Choose rules and number of controls based on the sigma performance of the test

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Quality control/2

Answers to Questions 56–59

- 56. D** QC is not required for diagnostic tests that are classified as waived tests under CLIA 1988 unless specified by the manufacturer or deemed accrediting agency. Subpart K specifies use of at least two levels of controls per 24 hours for all nonwaived tests (except for coagulation testing, which requires two levels every 8 hours, and blood gases, which requires three levels within 24 hours). eQC cannot be used to satisfy the requirements in subpart K as of January 1, 2016. An IQCP, if approved, permits laboratories to perform conventional QC with less frequency than required in subpart K.

57. **A** IQCPs must determine the risk of testing error to include preanalytical, analytical, and postanalytical errors and risk of harm resulting from such errors. These are applied to a matrix that is used to determine if the QC plan is at least equivalent to the CLIA standard. Generally, the laboratory must demonstrate that the frequency of device failure is unlikely and the severity of harm is mild when testing failure occurs.
58. **B** Sigma is calculated by subtracting bias from TE_a (as defined by CLIA 1988) and dividing the difference (which signifies the RE that can be tolerated) by the laboratory's standard deviation. The result determines the probability (odds) of reporting a laboratory result that is beyond the CLIA error limit. A laboratory test functioning at 2 sigma is expected to produce 308,537 errors per million tests (called defects per million opportunities [DPMO]). In this example, almost one in three laboratory results would be above or below 1.0 mg/dL of the expected value. Conversely, a test performing at 4 sigma would produce 6,210 erroneous results per million, and one functioning at 6 sigma would have just 3.4 outside the CLIA limit for acceptable performance.
59. **D** Tests that have very high reproducibility and low SE (bias) do not require as stringent QC limits as those that do not. Using the 4_{1s} and R_{4s} rules in such cases will result in rejecting tests that could have been reported because they fall within the CLIA range for allowable analytical error. In general, tests that perform above 5 sigma should not be rejected unless the 1_{3s} rule has been violated. In other words, results less than 3s from the mean are still within acceptable limits of error. For tests performing above 5 sigma, two controls measured every 1000 samples and at least once per day are adequate for detection of clinically significant errors, but tests performing at 5 sigma or less require more frequent controls. For tests performing at 5 sigma, it is advisable to include the 2_{2s} and R_{4s} rules using 2 controls per run and at least once per day, and for tests performing at 4 sigma or below the 4_{1s} rule should be added using 2 controls per run and 2 runs per day. For tests performing below 4 sigma use all of the conventional Westgard rules using 4 controls per run and 2 runs per day.

60. A new immunoassay for serum total cortisol is compared with a reference method, and the resultant data give a linear regression of $\hat{y} = 3.0 + 0.80x$. The new method's standard deviation is 1.5 µg/dL. The CLIA total error allowable (TE_a) is ±25% at any concentration. Given a decision level of 20 µg/dL, what is the total error estimate and would the new method be acceptable?

- A. TE_a = -1 µg/dL; acceptable
- B. TE_a = -3 µg/dL; acceptable
- C. TE_a = -4 µg/dL; acceptable
- D. TE_a = -5 µg/dL; unacceptable

Laboratory Operations/Evaluate laboratory data to assess validity/Accuracy of procedures/Laboratory Error/3

Answer to Question 60

- 60. C** The regression equation is used to estimate the SE between the methods. $\hat{y} = 3.0 + 0.8(20) = 19.0$ µg/dL. Bias = 19 - 20 = -1 µg/dL. On average, the new method

measures 1 µg/dL lower than the reference method at a concentration of 20 µg/dL, but there is considerable uncertainty as to what the actual difference would be. Because the standard deviation of the new method is 1.5, the 95% confidence interval for this result by the new method is $19.0 \pm 2(1.5) = 16.0 - 22.0$ µg/dL. The total error estimate is bias + (–2s) or $-1 + (-3) = -4$ µg/dL. The CLIA TEa is $\pm 25\%$ at any concentration. At 20 µg/dL, the TEa is 5 µg/dL. Therefore, the new method has an acceptable margin of error at this concentration based on the CLIA TEa limit for cortisol.

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